Smart Street Light: Wiring Guide (MB-102 breadboard + Arduino UNO)

All parts sit on the A-E side of the breadboard, so nothing crosses the middle gap. Holes in the same numbered row (A-E) are connected to each other.

Do not power the Arduino until every step is finished and checked.

Step 1: Power rails

From	To
Arduino 5V	Red (+) rail
Arduino GND	Blue (-) rail

If your rails are split in the middle, bridge the break with a jumper on both the red and blue rails.

Step 2: LED 1 (Arduino D5)

Part	Hole
Jumper from Arduino D5	A12
220 ohm resistor, leg 1	B12
220 ohm resistor, leg 2	B16
LED long leg (+)	C16
LED short leg (-)	C17
Jumper from blue (-) rail	A17

Path: D5 -> row 12 -> resistor -> row 16 -> LED -> row 17 -> GND

Step 3: LED 2 (Arduino D6)

Part	Hole
Jumper from Arduino D6	A22
220 ohm resistor, leg 1	B22
220 ohm resistor, leg 2	B26
LED long leg (+)	C26
LED short leg (-)	C27
Jumper from blue (-) rail	A27

Path: D6 -> row 22 -> resistor -> row 26 -> LED -> row 27 -> GND

Step 4: LDR with 10K resistor (Arduino A0)

Part	Hole
Jumper from red (+) rail	A32
LDR, leg 1	C32
LDR, leg 2	C34
Jumper to Arduino A0	A34
10K resistor, leg 1	B34
10K resistor, leg 2	B38
Jumper from blue (-) rail	A38

Path: 5V -> LDR -> row 34 (A0 reads here) -> 10K resistor -> GND

Brighter light gives a higher A0 value. Darkness gives a lower value.

Step 5: IR sensors (no breadboard holes needed)

Sensor	VCC	GND	OUT
IR 1	Red (+) rail	Blue (-) rail	Arduino D2
IR 2	Red (+) rail	Blue (-) rail	Arduino D3

Use male-to-female jumper wires from the sensor pins to the rails and the Arduino.

Pin summary

Arduino pin	Connected to
5V	Red (+) rail
GND	Blue (-) rail
D2	IR sensor 1 OUT
D3	IR sensor 2 OUT
D5	Row 12 (LED 1 via 220 ohm)
D6	Row 22 (LED 2 via 220 ohm)
A0	Row 34 (LDR / 10K junction)

Checks before powering on

1. LED long leg (+) is in the row connected to the resistor (rows 16 and 26). Short leg is in the row connected to GND (rows 17 and 27).
2. Resistor colours: 220 ohm is red-red-brown, 10K is brown-black-orange.
3. Every GND jumper goes to the blue (-) rail and every 5V jumper goes to the red (+) rail.
4. No two different wires share the same numbered row, except where the plan says they should.

Test LED 1 first

Build only Steps 1 and 2, then upload:

```
void setup() { pinMode(5, OUTPUT); }
void loop() { digitalWrite(5, HIGH); }
```

The LED should light steadily. If not: flip the LED, press every leg and jumper in firmly, or try another LED. Once it works, repeat for LED 2 on pin 6, then add the LDR and IR sensors, and upload

smart_street_light.ino.